

Unit IV

4

Tree Based and Probabilistic Models

4.1 : Tree Based Model : Decision Tree

Q.1 What is tree based model ?

Ans. : Tree-based models use a series of if-then rules to generate predictions from one or more decision trees. All tree-based models can be used for either regression (predicting numerical values) or classification (predicting categorical values)

Q.2 What is decision tree ? Explain.

Ans. : • A decision tree is a simple representation for classifying examples. A decision tree or a classification tree is a tree in which each internal node is labeled with an input feature. The arcs coming from a node labeled with a feature are labeled with each of the possible values of the feature. Each leaf of the tree is labeled with a class or a probability distribution over the classes.

- In this method a set of training examples is broken down into smaller and smaller subsets while at the same time an associated decision tree get incrementally developed. At the end of the learning process, a decision tree covering the training set is returned.
- The key idea is to use a decision tree to partition the data space into cluster (or dense) regions and empty (or sparse) regions.
- Decision tree consists of
 1. **Nodes :** Test for the value of a certain attribute.
 2. **Edges :** Correspond to the outcome of a test and connect to the next node or leaf.
 3. **Leaves :** Terminal nodes that predict the outcome.

(4 - 1)

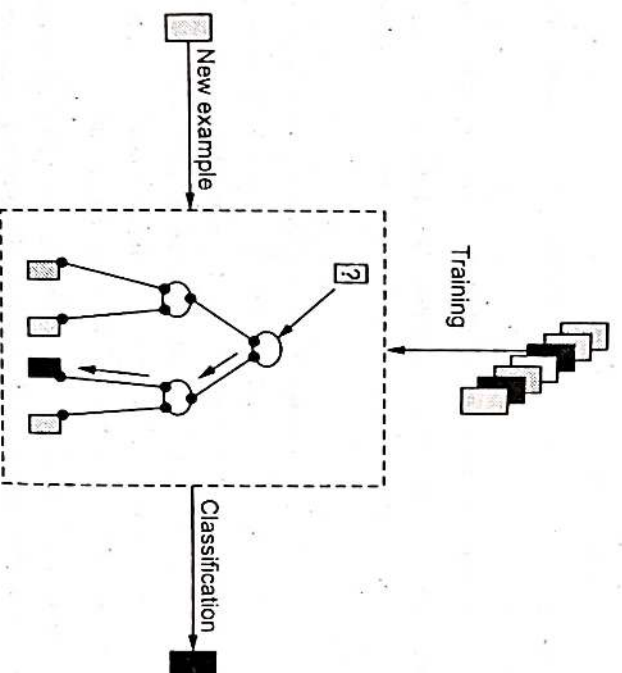


Fig. Q.2.1

- In Decision Tree Learning, a new example is classified by submitting it to a series of tests that determine the class label of the example. These tests are organized in a hierarchical structure called a decision tree.

Q.3 Write short on regression tree with its algorithm.

Ans. : • Regression tree models are known for their simplicity and efficiency, when dealing with domains with large number of variables and cases. Regression trees are obtained using a fast divide and conquer greedy algorithm that recursively partitions the given training data into smaller subsets.

- When the complexity of the model is dependant on the learning sample size, both bias and variance decrease with the learning sample size. E.g. regression trees. Small bias, a tree can approximate any non linear function.
- Regression trees are among the machine learning method that present the highest variance. Even a small change of the learning sample can result in a very different tree. Even small trees have a high variance.

- Possible sources of variance :

1. Discretization of numerical attributes : The selected threshold has a high variance.
2. Structure choice : Sometimes, attribute scores are very close.
3. Estimation at leaf nodes : Because of the recursive partitioning, prediction at leaf nodes are based on very small samples of objects.

• Regression trees are constructed using a Recursive Partitioning (RP) algorithm. This algorithm builds a tree by recursively splitting the training sample into smaller subsets. The RP algorithm receives as input a set of n data points and if certain termination criteria are not met it generates a test node t , whose branches are obtained by applying the same algorithm with two subsets of the input data points.

- At each node the best split test is chosen according to some local criterion which means that this is a greedy hill-climbing algorithm.

Algorithm : Recursive Partitioning Algorithm

Input : A set of n datapoints

Output : A regression tree

IF termination criterion THEN

Create Leaf Node and assign it a Constant Value

Return Leaf Node

ELSE

Find Best Splitting Test s^*

Create Node t with s^*

Left_branch (t) = RecursivePartitioningAlgorithm
($\{ < x_i, y_i > : x_i \rightarrow s^* \}$)

Right_branch (t) = RecursivePartitioningAlgorithm
($\{ < x_i, y_i > : x_i \rightarrow s^* \}$)

Return Node t

ENDIF

- The algorithm has three main components :

 1. A way to select a split test

2. A rule to determine when a tree node is terminal.
3. A rule for assigning a value to each terminal node.

Q.4 Explain advantages and disadvantages of decision tree.

Ans. : Advantages :

1. Rules are simple and easy to understand.
2. Decision trees can handle both nominal and numerical attributes.
3. Decision trees are capable of handling datasets that may have errors.
4. Decision trees are capable of handling datasets that may have missing values.
5. Decision trees are considered to be a nonparametric method.
6. Decision trees are self-explanatory.

Disadvantages :

1. Most of the algorithms require that the target attribute will have only discrete values.
2. Some problem are difficult to solve like XOR.
3. Decision trees are less appropriate for estimation tasks where the goal is to predict the value of a continuous attribute.
4. Decision trees are prone to errors in classification problems with many class and relatively small number of training examples.

Q.5 Define feature tree.

Ans. : Feature tree is a tree such that each internal node is labelled with a feature, and each edge emanating from an internal node is labelled with a literal. The set of literals at a node is called a split. Each leaf of the tree represents a logical expression, which is the conjunction of literals encountered on the path from the root of the tree to the leaf.

Q.6 How empirical probabilities can be used in ranking and probability estimation trees ? Explain the purpose of pruning the subtree of a decision tree.

[SPPU : Dec-17, End Sem, Marks 8]

Ans. :

- The class distributions in the leaves of an unlabelled feature tree can be used to turn one and the same tree into a decision tree, a ranking tree, or a probability estimation tree :

- a) To turn a feature tree into a ranker, we order its leaves on non-increasing empirical probabilities, which is provably optimal on the training set;
 - b) To turn the tree into a probability estimator, we predict the empirical probabilities in each leaf, applying Laplace or m-estimate smoothing to make these estimates more robust for small leaves;
 - c) To turn the tree into a classifier, we choose the operating conditions and find the operating point that is optimal under those operating conditions.
- A tree is defined as a set of logical conditions on attributes ; a leaf represents the subset of instances corresponding to the conjunction of conditions along its branch or path back to the root. A simple approach to ranking is to estimate the probability of an instance's membership in a class and assign that probability as the instance's rank. A decision tree can easily be used to estimate these probabilities.
 - Rule learning is known for its descriptive and therefore comprehensible classification models which also yield good class predictions. In some application areas, we also need good class probability estimates.
 - A probabilistic rule is an extension of a classification rule, which does not only predict a single class value, but a set of class probabilities, which form a probability distribution over the classes. This probability distribution estimates all probabilities that a covered instance belongs to any of the class in the data set, so we get one class probability per class.
 - Building decision trees with accurate probability estimates, called probability estimation trees. A small tree has a small number of leaves, thus more examples will have the same class probability. That prevents the learning algorithm from building an accurate PET.
 - Purpose of pruning the subtree of a decision tree.
 - Pruning simplifies a classifier by merging disjuncts that are adjacent in instance space. This can improve the classifier's performance by eliminating error-prone components.

- Pruning of the decision tree is done by replacing a whole sub-tree by a leaf node. The replacement takes place if a decision rule establishes that the expected error rate in the sub-tree is greater than in the single leaf.

Q.7 Write the Grow Tree algorithm to generate feature tree? Explain the role of best split in this algorithm.

IES [SPPU : May - 16, End Sem, Marks 9]

Ans. : • A feature tree is a tree such that each internal node is labelled with a feature, and each edge emanating from an internal node is labelled with a literal. The set of literals at a node is called a split.

- Each leaf of the tree represents a logical expression, which is the conjunction of literals encountered on the path from the root of the tree to the leaf. The extension of that conjunction is called the instance space segment associated with the leaf.

• Algorithm GrowTree(D, F)

Input : data D; set of features F .

Output : feature tree T with labelled leaves.

1. if Homogeneous(D) then return Label(D);
 2. $S \leftarrow \text{BestSplit}(D, F)$;
 3. split D into subsets D_i according to the literals in S ;
 4. for each i do
 5. if $D_i \neq \emptyset$; then $T_i \leftarrow \text{GrowTree}(D_i, F)$;
 6. else T_i is a leaf labelled with Label(D);
 7. end
 8. return a tree whose root is labeled with S and whose children are T_i .
- Algorithm gives the generic learning procedure common to most tree learners. It assumes that the following three functions are defined :
 1. Homogeneous(D) returns true if the instances in D are homogeneous enough to be labelled with a single label, and false otherwise;
 2. Label(D) returns the most appropriate label for a set of instances D;
 3. BestSplit(D, F) returns the best set of literals to be put at the root of the tree.

- a) To turn a feature tree into a ranker, we order its leaves on non-increasing empirical probabilities, which is provably optimal on the training set;
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- These functions depend on the task at hand : For instance, for classification tasks a set of instances is homogeneous if they are of a single class, and the most appropriate label would be the majority class. For clustering tasks a set of instances is homogenous if they are close together, and the most appropriate label would be some exemplar such as the mean.

Q.8 If S is a collection of 14 examples with 9 YES and 5 NO examples then calculate entropy.

Ans. :

$$\text{Entropy}(S) = \sum p(i) \log_2 p(i)$$

Where $p(i)$ is the proportion of S belonging to class I.

Σ is over c.

$$\text{Entropy}(S) = - \left(\frac{9}{14} \log_2 \left(\frac{9}{14} \right) - \left(\frac{5}{14} \log_2 \left(\frac{5}{14} \right) \right) \right) = -0.940$$

Q.9 Consider following splits having four features :

Length = [3, 4, 5] [2+, 0-] [1+, 3-] [2+, 2-]

Gills = [Yes, No] [0+, 4-] [5+, 1-]

Beak = [Yes, No] [5+, 3-] [0+, 2-]

Teeth = [many, few] [3+, 4-] [2+, 1-]

Find : Total weighted entropy and gini-index of all features.

[SPPU : Dec-18, In Sem, Marks 8]

Ans. : • Lets calculate the impurity of the first split. We have three segments : the first one is pure and so has entropy 0;

- The second one has entropy

$$= -(1/4) \log_2 (1/4) \log_2 -(3/4) \log_2 (3/4) = 0.5 + 0.31$$

$$= 0.81 ; \text{ the third one has entropy}$$

- Similar calculations for the other three features give the following entropies :

$$\text{Gills} = (4/10) \times 0 + (6/10) \times (-5/6) \log_2 (5/6) - (1/6) \log_2 (1/6) = 0.39$$

$$\text{Beak} = (8/10) \times (-5/8) \log_2 (5/8) - (3/8) \log_2 (3/8) + (2/10) \times 0 = 0.76$$

$$\text{Teeth} = (7/10) \times (-3/7) \log_2 (3/7) - (4/7) \log_2 (4/7)$$

$$+ (3 \times 10) \times (-2/3) \log_2 (2/3) - (1/3) \log_2 (1/3) = 0.97$$

We thus clearly see that 'Gills' is an excellent feature to split on ; 'Teeth' is poor and the other two are somewhere in between.

4.2 : Probabilistic Model

Q.10 What is a probabilistic model ? Give an example of it.

[SPPU : Dec-17, End Sem, Marks 4]

Ans. : • Probabilistic modeling is a statistical technique used to take into account the impact of random events or actions in predicting the potential occurrence of future outcomes.

- In machine learning, we train the system by using a limited data set called 'training data' and based on the confidence level of the training data we expect the machine learning algorithm to depict the behaviour of the larger set of actual data.

- Probability theory provides a mathematical foundation for quantifying uncertainty of the knowledge.

- ML is focused on making predictions as accurate as possible, while traditional statistical models are aimed at inferring relationships between variables.

- We make observations using the sensors in the world. Based on the observations, we intend to make decisions. Given the same observations, the decision should be the same. However, the world changes, observations change, our sensors change, the output should not change.

- We build models for predictions; can we trust them? Are they certain? Many applications of machine learning depend on good estimation of the uncertainty :

- Forecasting
- Decision making
- Learning from limited, noisy, and missing data
- Learning complex personalised models
- Data compression
- Automating scientific modelling, discovery, and experiment design

- A signal is called random if its occurrence can not be predicted. Such signal can not be by any mathematical equation.
- The random signals are represented collectively by a random variable takes its value will be taken at particular time is not known.

Q.11 What is Bayes theorem ? Explain.

Ans. : • Bayes' theorem is a method to revise the probability of an event given additional information. Bayes's theorem calculates a conditional probability called a posterior or revised probability.

• Bayes' theorem is a result in probability theory that relates conditional probabilities. If A and B denote two events, $P(A|B)$ denotes the conditional probability of A occurring, given that B occurs. The two conditional probabilities $P(A|B)$ and $P(B|A)$ are in general different.

• Bayes theorem gives a relation between $P(A|B)$ and $P(B|A)$. An important application of Bayes' theorem is that it gives a rule how to update or revise the strengths of evidence-based beliefs in light of new evidence a posteriori.

• A prior probability is an initial probability value originally obtained before any additional information is obtained.

• A posterior probability is a probability value that has been revised by using additional information that is later obtained.

• Suppose that $B_1, B_2, B_3 \dots B_n$ partition the outcomes of an experiment and that A is another event. For any number, k, with $1 \leq k \leq n$, we have the formula :

$$P(B_k/A) = \frac{P(A/B_k) \cdot P(B_k)}{\sum_{i=1}^n P(A/B_i) \cdot P(B_i)}$$

Q.12 Write short note on class probability estimation.

Ans [SPPU : May -16, End Sem, Marks 8, Oct-16, In Sem, Marks 5]

Ans. : • Probability estimates are also important when the classification outputs are not used in isolation but are combined with information from other components in a system.

• Let us consider handwritten character recognition example. Here the output from the classifier is used as input to a high level system which incorporates domain information.

• For many supervised learning tasks it is very costly to produce training data with class labels.

• Active learning acquires data incrementally, at each stage using the model learned so far to help identify especially useful additional data for labeling.

• Many applications require more than simple classification. Decision making often requires estimates of the probability of class membership.

• Class probability estimates can be combined with decision making costs/benefits to minimize expected cost.

• For example, in target marketing the estimated probability that a customer will respond to an offer is combined with the estimated profit. Other applications require ranking of cases, to add flexibility to user processing.

• Given the observation x and the class label y, we assume that the estimated pair-wise class probabilities μ_{ij} of $\mu_{ij} = p(y = i | x = j, x)$ are available.

• From the i^{th} and j^{th} classes of a training set, we obtain a model which, for any new x, calculates μ_{ij} as an approximation of μ_{ij} .

• Then, using all μ_{ij} , the goal is to estimate $P_i = p(y = i | x)$, for $i = 1, \dots, k$.

Q.13 Write and explain Naive Bayes classification algorithm.

Ans [SPPU, May-18, End Sem, Marks 8]

OR Explain Naive Bayes classification algorithm.

Ans [SPPU, May-17, End Sem, Marks 8]

OR Is Naive Bayes algorithm supervised or unsupervised task ? Explain how it achieves the task you specified.

Ans [SPPU : Dec-16, End Sem, Marks 8]

Ans. : • Naive Bayes algorithm is a supervised learning algorithm, which is based on Bayes theorem and used for solving classification problems.

• Naive Bayes assumes conditional independence over the training dataset. The classifier separates data into different classes according to the Bayes' Theorem. But assumes that the relationship between all input features in a class is independent. Hence, the model is called Naive.

• Essentially, the Naive Bayes model is a conditional probability classification with Bayes Theorem applied.

• Conditional probability defines the probability of an event occurring given the occurrence of another event. The conditional probability is used to calculate the joint probability, which is the probability of two or more simultaneous events. On the flip side, the joint probability could also be used to calculate the conditional probability, but it's often quite difficult to calculate the joint probability hence we use Bayes Theorem to calculate the conditional probability.

• A Naive Bayes classifier is a program which predicts a class value given a set of attributes.

• For each known class value,

1. Calculate probabilities for each attribute, conditional on the class value.
2. Use the product rule to obtain a joint conditional probability for the attributes.
3. Use Bayes rule to derive conditional probabilities for the class variable.

• Once this has been done for all class values, output the class with the highest probability.

• Naive Bayes simplifies the calculation of probabilities by assuming that the probability of each attribute belonging to a given class value is independent of all other attributes. This is a strong assumption but results in a fast and effective method.

• The probability of a class value given a value of an attribute is called the conditional probability. By multiplying the conditional probabilities

together for each attribute for a given class value, we have a probability of a data instance belonging to that class.

Q.14 At a certain university, 4 % men are over 6 feet tall and 1 % women are over 6 feet tall. The total student population is divided in the ratio 3 : 2 in favor of women. If a student is selected at random from among all those over six feet tall, what is the probability that the student is a woman ? [SPPU : Dec.-19, End Sem, Marks 8]

Ans. : • Let M be the event that student is male and F be the event that the student is female.

• Let T be the event that student is taller than 6 ft.

$$P(M) = 2/5$$

$$P(F) = 3/5$$

$$P(T|M) = 4/100$$

$$P(T|F) = 1/100$$

$$\begin{aligned} P(F/T) &= \frac{P(T/F) \times P(F)}{(P(T/F) \times P(F) + P(T/M) \times P(M))} \\ &= \frac{(1/100) \times (3/5)}{(1/100) \times (3/5) + (4/100) \times (2/5)} = \frac{3}{11} \end{aligned}$$

Q.15 Consider following dataset and predict the class of new instance X using Naive Bayes classification algorithm.

Tid	Refund	Marital Status	Taxable Amount	Evade
1	Yes	Single	125 K	No
2	No	Married	100 K	No
3	No	Single	70 K	No
4	Yes	Married	120 K	No
5	No	Divorced	95 K	Yes
6	No	Married	60 K	No
7	Yes	Married	220 K	No

8	No	Single	85 K	Yes
9	No	Married	75 K	No
10	No	Single	90 K	Yes

$X = (\text{Refund} = \text{No}, \text{Marital status} = \text{Married}, \text{Income} = 120 \text{ K})$

[SPPU : June-19, End Sem, Marks 10]

Ans. :

$$P(\text{Refund} = \text{Yes}|\text{No}) = 3/7$$

$$P(\text{Refund} = \text{No}|\text{No}) = 4/7$$

$$P(\text{Refund} = \text{Yes}|\text{Yes}) = 0$$

$$P(\text{Refund} = \text{No}|\text{Yes}) = 1$$

$$P(\text{Marital Status} = \text{Single}|\text{No}) = 2/7$$

$$P(\text{Marital Status} = \text{Divorced}|\text{No}) = 1/7$$

$$P(\text{Marital Status} = \text{Married}|\text{No}) = 4/7$$

$$P(\text{Marital Status} = \text{Single}|\text{Yes}) = 2/7$$

$$P(\text{Marital Status} = \text{Divorced}|\text{Yes}) = 1/7$$

$$P(\text{Marital Status} = \text{Married}|\text{Yes}) = 0$$

For taxable income :

If class = No : sample mean = 110

sample variance = 2975

If class = Yes: sample mean = 90

sample variance = 25

$$P(X|\text{Class} = \text{No}) = P(\text{Refund} = \text{No}|\text{Class} = \text{No}) \times P(\text{Married}|\text{Class} = \text{No}) \\ \times P(\text{Income} = 120 \text{ K}|\text{Class} = \text{No})$$

$$= 4/7 \times 4/7 \times 0.0072 = 0.0024$$

$$P(X|\text{Class} = \text{Yes}) = P(\text{Refund} = \text{No}|\text{Class} = \text{Yes}) \times P(\text{Married}|\text{Class} = \text{Yes}) \\ \times P(\text{Income} = 120 \text{ K}|\text{Class} = \text{Yes})$$

$$= 1 \times 0 \times 1.2 \times 10^{-9} = 0$$

Since

$$P(X|\text{No})P(\text{No}) > P(X|\text{Yes})P(\text{Yes})$$

Therefore $P(\text{No}|X) > P(\text{Yes}|X) \Rightarrow \text{Class} = \text{No}$

Q.16 Consider following dataset and predict the class of new instance X using Naive Bayes.

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Classification algorithm.

$X = (\text{Outlook} = \text{Sunny}, \text{Temp.} = \text{Cool}, \text{Humidity} = \text{High},$

$\text{Wind} = \text{Strong}).$

[SPPU : May-19, End Sem, Marks 10]

Ans. : Learning phase

Outlook	Play=Yes	Play=No	Temperature	Play=Yes	Play=No
Sunny	2/9	3/5	Hot	2/9	2/5